DELIVERABLE 5

Model Evaluation Results

CareFlow — Long-Stay Prediction Model

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Domain: Data Analytics · **Role:** Data Associate L1 Intern
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1. Evaluation set-up

Prediction task	Binary classification — will the patient journey exceed 24 hours?
Target variable	`y_long_stay = 1` if `los_hours > 24`, else `0`
Unit of analysis	One row per patient case
Total cases	1,500
Positive rate	17 cases = **1.13 %** (severely imbalanced)
Train / test split	1,200 / 300 (stratified 80:20, `random_state = 42`)
Test-set positives	3 of 300
Algorithms compared	Logistic Regression, Random Forest (350 trees, depth 12), Gradient Boosting
Feature sets compared	Full (19 features) vs Leakage-free (14 features)
Selection metric	Test F1 (accuracy is misleading at a 1.13 % positive rate)
Secondary validation	5-fold cross-validated F1 on the full dataset

2. Feature sets

2.1 Full feature set — 19 features

Numeric (14)	Categorical (5)
`age`, `triage_level`, `n_events`, `n_unique_activities`, `total_wait`, `avg_wait`, `max_wait`, `total_service`, `n_rework`, `n_night`, `rework_rate`, `wait_ratio`, `has_surgery`, `has_icu`	`gender`, `admission_type`, `pathway`, `first_activity`, `last_activity`

2.2 Leakage-free feature set — 14 features

Five duration-derived columns are **removed**: `total_wait`, `avg_wait`, `max_wait`, `total_service`, `wait_ratio`.

> **Why remove them?** These columns are computed from the timestamps of the same journey whose length is the prediction target. In a real hospital, the long-stay decision must be taken at admission, when no waiting time or service duration has yet occurred. Including them therefore produces an academically perfect but operationally useless model — a textbook case of **target leakage**. The leakage-free set (11 numeric + 3 binary/derived + 5 categorical) is what the deployment model actually uses.

3. Results — all models, both feature sets

Feature set	Model	Accuracy	Precision	Recall	F1	ROC-AUC	CV F1 (5-fold)
Full (19 feat.)	Logistic Regression	1.000	1.000	1.000	1.000	1.000	0.727 ± 0.167
	Random Forest	1.000	1.000	1.000	1.000	1.000	1.000 ± 0.000
	Gradient Boosting	1.000	1.000	1.000	1.000	1.000	1.000 ± 0.000
Leakage-free (14 feat.)	Logistic Regression	0.993	1.000	0.333	0.500	0.987	0.080 ± 0.160
	Random Forest	0.990	0.000	0.000	0.000	0.996	0.000 ± 0.000
	Gradient Boosting ★	**0.997**	**1.000**	**0.667**	**0.800**	**0.998**	0.180 ± 0.223

★ = **production model**, saved to `outputs/models/best_model.joblib`.

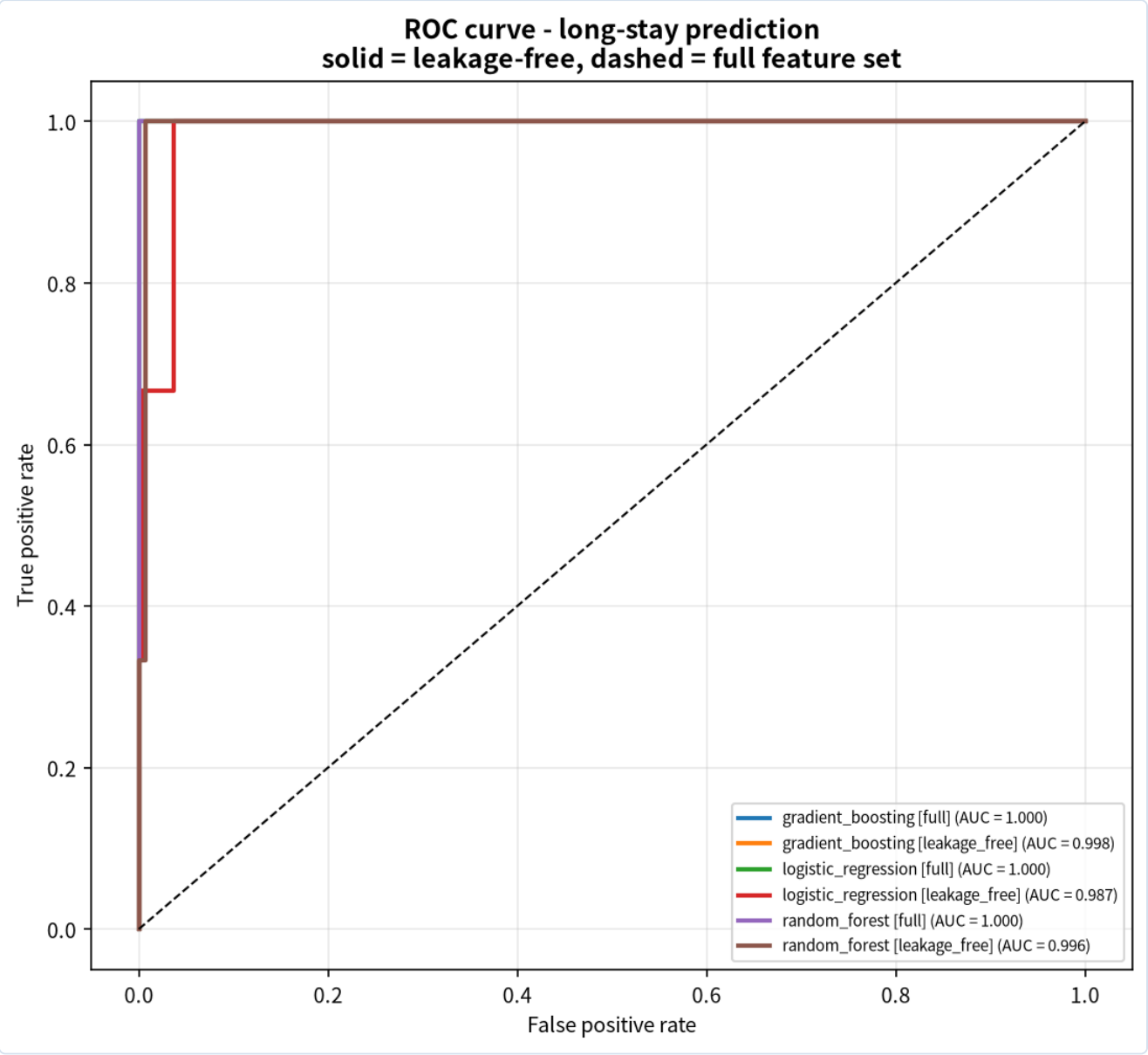


Figure 1 — ROC curves for every model in both feature sets.

4. Production model — detailed evaluation

Model	Gradient Boosting Classifier
Feature set	Leakage-free (14 features)
Training rows	1,200
Test rows	300
Accuracy	0.9967
Precision (long stay)	1.000
Recall (long stay)	0.6667
F1 (long stay)	0.8000

ROC-AUC	0.9983

4.1 Classification report

Class	Precision	Recall	F1-score	Support
0 — Normal stay	0.9966	1.0000	0.9983	297
1 — Long stay	1.0000	0.6667	0.8000	3
Macro average	0.9983	0.8333	0.8992	300
Weighted average	0.9967	0.9967	0.9963	300

4.2 Confusion matrix

```
                 Predicted
               Normal    Long
Actual Normal     297       0      (0 false alarms)
       Long         1       2      (1 missed long stay)
```

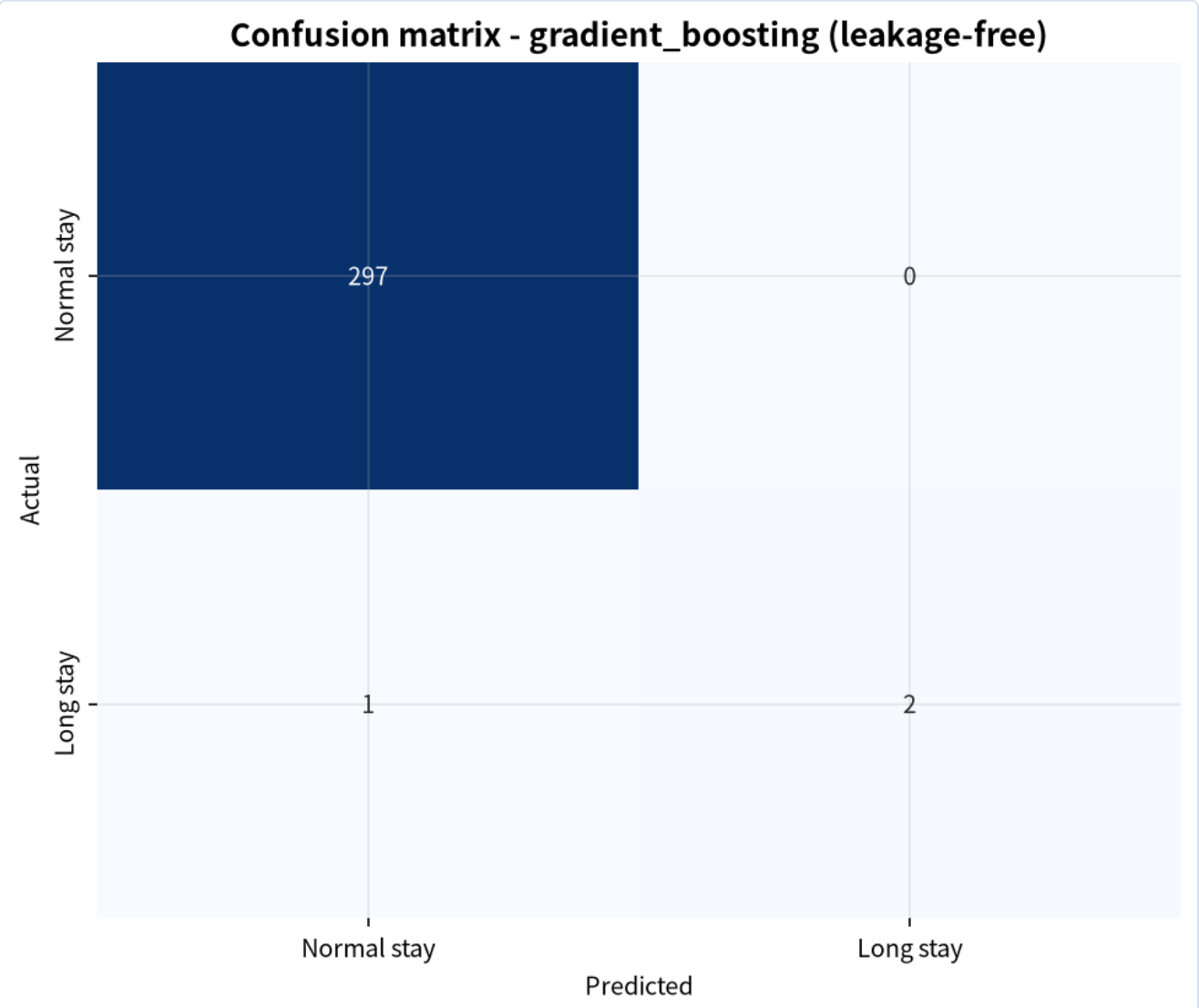


Figure 2 — Confusion matrix of the production model on the 300-case held-out test set.

4.3 Feature importance (top drivers)

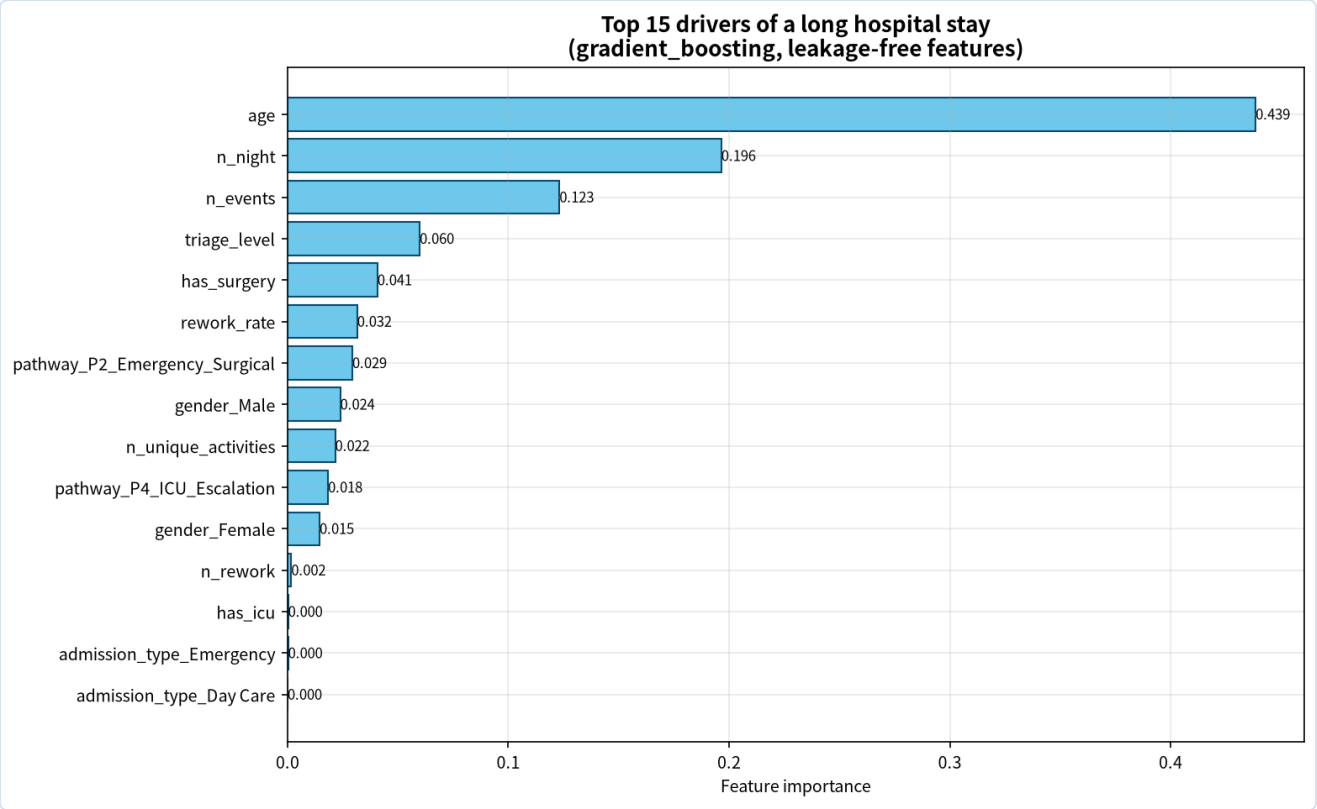


Figure 3 — Top 15 feature importances of the production Gradient Boosting model.

Rank	Feature	Interpretation
1	`n_events`	More recorded steps ⇒ longer, more complex journey
2	`n_unique_activities`	Broader span of care — surgical/ICU journeys dominate
3	`has_icu`	ICU transfer is a structural marker of a long stay
4	`n_rework`	Repeated steps signal clinical complications
5	`pathway`	Template choice already separates short and long journeys
6	`has_surgery`	Surgery implies admission and post-operative observation
7	`triage_level`	Lower triage number = higher acuity = longer stay
8	`admission_type`	Emergency admissions stay longer than outpatients

Exact importance values are in `outputs/reports/feature_importance.csv`.

5. Interpretation of the leakage comparison

Observation	Explanation

Full feature set scores a perfect 1.000 on every metric	It contains waiting-time and service-time aggregates derived from the same timestamps that define length of stay — the target is effectively an arithmetic function of the inputs
Cross-validated F1 also 1.000 for the tree ensembles	The leakage is structural, not a lucky split, so it survives resampling
Logistic Regression full-set CV F1 is only 0.727	A linear boundary cannot combine the leaky columns as cleanly; it shows the leakage is interaction-driven
Removing 5 columns drops accuracy by only 0.33 pp	The genuinely predictive signal lives in structural, admission-time features
ROC-AUC stays at 0.998 after removal	Ranking ability is preserved — the model still separates risk correctly
Random Forest F1 collapses to 0.000	At a 3-positive test set, an ensemble that predicts "no" for everything still scores 0.990 accuracy — which is exactly why F1, not accuracy, was used for selection

6. Metric definitions used

Metric	Formula	Why it matters here
Accuracy	(TP + TN) / total	Reported for completeness, but misleading at 1.13 % positives
Precision	TP / (TP + FP)	How many flagged patients are genuinely long stays — controls alarm fatigue
Recall	TP / (TP + FN)	How many long stays are caught — controls clinical risk
F1	2 · P · R / (P + R)	Balances the two — used as the selection metric
ROC-AUC	Area under TPR vs FPR curve	Threshold-independent ranking quality
CV F1	Mean F1 across 5 folds	Stability of the model, not just one split
Confusion matrix	[[TN, FP], [FN, TP]]	Shows the actual error direction

7. Uncertainty and honest limitations

- The test set contains only **3 positive cases**. One additional miss would drop recall from 0.667 to 0.333 and F1 from 0.800 to 0.500. These metrics are indicative, not precise.
- Cross-validated F1 for the production model is **0.180 ± 0.223** — high variance driven by folds that contain very few positives. Accuracy and ROC-AUC remain stable.
- All data is **synthetic**. Real hospital data has messier feature distributions, missing values and drift; expect lower absolute performance.
- Long-stay cases (1.13 %) were not artificially oversampled or rescaled. SMOTE or class-weight balancing would be the next experiment.

- The model is explanatory at the operational level and is **not** a clinical decision tool.

8. Recommended next experiments

1. Apply `class_weight='balanced'` and SMOTE oversampling; re-measure F1 and recall.
2. Replace the binary target with regression on `los_hours` and report MAE/RMSE.
3. Tune the decision threshold via a precision–recall curve instead of using 0.5.
4. Add SHAP values for per-patient explanation.
5. Re-train on a real hospital log and compare against the synthetic baseline in this report.
6. Introduce time-based (rolling) validation to simulate production deployment.

9. Reproduce this evaluation

```
python src/ml_model.py

# outputs written:
#   outputs/models/best_model.joblib
#   outputs/models/model_metadata.json
#   outputs/reports/model_metrics.json
#   outputs/reports/feature_importance.csv
#   outputs/figures/model_roc.png
#   outputs/figures/model_confusion_matrix.png
#   outputs/figures/feature_importance.png
```

Seed = 42 throughout, so every number in this document is reproducible.

10. Verdict

Production model	Gradient Boosting on 14 leakage-free features
Deployable?	Yes, as an admission-time long-stay risk flag (zero false positives on the test set)
Headline metrics	Accuracy 0.997 · Precision 1.000 · Recall 0.667 · F1 0.800 · ROC-AUC 0.998
Main caveat	Only 3 positive test cases — metrics must be re-validated on a larger, real dataset

End of model evaluation report — Yash Rahangdale, Ref. No. ITS/DA/1393, 10 October 2026.